Fundmentals 28

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Problem 28.1

Let $\{a, b, c\} = X$. Since there are only three possible values for the size of the smallest element in a filter, there are three filters on X up to relabeling:

1. $\{X\}$
2. $\{\{a, b\}, X\}$
3. $\{\{a\}, \{a, b\}, X\}$

Problem 28.2

We prove the three axioms of a filter:

1. $\mathcal{B}$ only contains nonempty subsets of X so $\varnothing \notin \mathcal{F}$.
2. Let $F_1, F_2 \in \mathcal{F}$. We know that $F_1 \supseteq B_1$ and $F_2 \supseteq B_2$ where $B_1, B_2 \in \mathcal{B}$, so $F_1 \cap F_2 \supseteq B_1 \cap B_2 \supseteq B_3$ for some $B_3 \in \mathcal{B}$. Since $F_1 \cap F_2$ is a superset of an element of $\mathcal{B}$, $F_1 \cap F_2 \in \mathcal{F}$.
3. Since $\mathcal{F}$ is generated by $\mathcal{B}$, it is the set of all supersets of $\mathcal{B}$, so every $F \in \mathcal{F}$ has $G \in \mathcal{F}$ where $F \subseteq G$.

Problem 28.3

1. $\varnothing \not\supseteq A$, so $\varnothing \notin \mathcal{F}$.
2. Let $F, G \in \mathcal{F}$. Then $F \cap G \supseteq A$, so $F \cap G \in \mathcal{F}$.
3. Let $F \in \mathcal{F}$ and $F \subseteq G$. Then $G \supseteq F \supseteq A$ so $G \in \mathcal{F}$.

Problem 28.5

Let $B_1 = f(F_1)$ and $B_2 = f(F_2)$ for any $F_1, F_2 \in \mathcal{F}$. We know that $F_1 \cap F_2 \in \mathcal{F}$, so $f(F_1 \cap F_2) \in \mathcal{B}$. We have that $f(F_1 \cap F_2) \subseteq f(F_1) \cap f(F_2) = B_1 \cap B_2$. So we can let $B_3 = f(F_1 \cap F_2) \in \mathcal{B}$, so that $B_3 \subseteq B_1 \cap B_2$.

Problem 28.6

Assume there is finer $\mathcal{G} \supseteq \mathcal{F}$ with $\mathcal{G} \to x$. Since $F \subseteq G$, every open neighborhood of x is in $\mathcal{G}$ and therefore $\mathcal{F}$ as well. So x is a limit point of $\mathcal{F}$.

Now assume that x is a limit point of $\mathcal{F}$, so that $x \in \bigcap_{F \in \mathcal{F}} \overline{F}$. Let $\mathcal{B} = \{U \cap F : F \in \mathcal{F},\ U \text{ open neighborhood of } x\}$.

Let U be an open neighborhood of x. Then $U \supseteq U \cap F \in \mathcal{F}$, so $U \in \mathcal{G}$. Also, $\mathcal{G} \supseteq \mathcal{F}$ because, if we let $F \in \mathcal{F}$, then $U \cap F \subseteq F$ and $U \cap F \in \mathcal{G}$ so $F \in \mathcal{G}$. Therefore $G \to x$.

Problem 28.8

Let $\mathcal{F}$ be an ultrafilter and $\mathcal{G} \supseteq \mathcal{F}$. For contradiction, assume there's $G \in \mathcal{G} \backslash \mathcal{F}$. Since $G \notin \mathcal{F}$, $X \backslash G \in \mathcal{F}$, so $X \backslash G \in \mathcal{G}$. But then $\mathcal{G}$ contains G and $X \backslash G$, which is impossible since $\mathcal{G}$ is a filter. So we have a contradiction, and $\mathcal{F}$ is the finest filter on X.